

TechCorp Platform Architecture

This document describes the reference architecture used by TechCorp engineering teams. TechCorp is based in Bangalore and builds data-intensive products for enterprise customers.

The platform is organised around event streaming, caching, relational storage and graph analytics. Each project selects components from this reference architecture.

Event Streaming with Kafka

Kafka is a distributed event streaming platform used for high-throughput, fault-tolerant data pipelines. Kafka stores events in partitioned, replicated topics so that producers and consumers are decoupled.

At TechCorp, Kafka is the backbone for asynchronous communication between services. Consumer groups allow several instances of a service to share the load of a topic.

Project Alpha Architecture

Rahul manages Project Alpha. Project Alpha is the real-time order analytics platform of TechCorp.

Project Alpha uses Kafka, Redis and PostgreSQL. The ingestion service of Project Alpha is built with FastAPI.

Redis is used in Project Alpha as a low-latency cache for aggregated metrics, while PostgreSQL stores the curated order history.

Project Gamma Architecture

Priya manages Project Gamma. Project Gamma is the supplier knowledge graph initiative.

Project Gamma uses Kafka, Neo4j and FastAPI. Project Gamma depends on Project Alpha for order events.

Neo4j is a native graph database that stores data as nodes and relationships and is queried with Cypher.